# TUAN QUANG

AI Engineer | Machine Learning Model | AI-Driven Solutions
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## Summary

AI professional with over 5 years of experience in software engineering, multi-agent, AI-driven solutions, and MLOps. Seeking a full-time AI Engineer position where I can leverage my skills in designing scalable AI solutions, and building cloud-native MLOps pipelines to advance your mission of transforming core workflows through AI-powered intelligence.

## Experience

### Galileo Financial Technologies — Remote
Jr Software AI Engineer — 04/2025 - 09/2025

Financial Company, applying AI and ML in a financial system

- Architected end-to-end MLOps infrastructure on AWS utilizing Docker, SageMaker, and GitHub Actions, automating model deployment and experiment tracking via Weights & Biases and DVC that can reduce the operation cost to 25%.
- Optimized LLM inference and data throughput by 15% by architecting scalable backend APIs and implementing advanced log analysis with Splunk, ensuring 99.9% uptime for high-frequency financial integrations.
- Built Python-based ML services on Amazon Bedrock and SageMaker, fine-tuning models on financial data. Optimized strategies improved fraud detection by 40%.

### LPL Financial — Remote
Software Engineer — 01/2024 - 12/2024

Leading wealth management firm for independent advisors across the US

- Architected an end-to-end **agentic RAG system** with short- and long-term memory using **LangGraph, LangChain, and CrewAI**, leveraging **Weaviate** for **vector–graph hybrid search** to cut manual document research time by 70%.
- Engineered high-performance **inference pipelines** on **AWS SageMaker** and **Bedrock**, fine-tuning Claude 3.5, Llama 3 (70B), and Gemma 3, and applying **model quantization** to automate complex domain tasks and reduce operational latency by 25%.
- Deployed cloud-native infrastructure on **AWS (Lambda, S3, DynamoDB, EC2, ECR, ECS)** with **Terraform, Docker, Nginx, PostgreSQL** (indexed/optimized SQL), and full observability via **OpenTelemetry, Grafana, and Prometheus** to ensure 99.9% reliable, **large-scale AI microservices.**

### Ford Credit — Remote
Software Engineer — 02/2022 - 12/2023

Financial services arm of Ford Motor Company

- Designed high-availability **backend microservices on GCP/PCF** using **Java**, **Spring Boot,** and resilient **PostgreSQL** schemas, while modernizing the **Angular + Material UI** frontend to deliver a responsive dashboard that improved experiment data accessibility by 75%.
- Automated multi-environment infrastructure with **Terraform** and Tekton **CI/CD**, ensuring 99.9% uptime via rigorous testing.
- Built an interactive dashboard using advanced SQL techniques to track business requirements and provide **real-time visibility into key performance metric.**

### AeroVironment — Simi Valley, United States
Software Engineer — 01/2020 - 02/2022

Global leader in intelligent, multi-domain robotic systems

- Achieved a 25% reduction in mission execution time by **architecting a production-ready autonomous flight control system** for UAVs with optimized routing and decision logic.
- Built a **cross-platform** control interface using **C# (.NET)** for backend services and **React/JavaScrip**t for the UI, delivering consistent performance and feature parity across **Windows, Android, and Linux**.
- Developed low-latency communication layers in C# for real-time telemetry. Enhanced flight-path visualization and situational awareness.

## Skills

AI & Deep Learning:
PyTorch, TensorFlow, JAX, ONNX, Hugging Face (Transformers, Accelerate, Diffusers), Scikit-learn, OpenCV, Ollama, LangGraph, CrewAI, AutoGen, LangChain, LlamaIndex, RagFlow, RAG, LLM Fine-tuning, OpenAI & Claude SDKs, WandB, FAISS, Pinecone, Chroma, NumPy, MLOps, Pandas, Detectron2

Software Engineering:
Python (Proficient), Java, JavaScript, TypeScript, SQL, C++, C#, R, C, Objective-C, Assembly, FastAPI, Spring Boot, Node.js, REST APIs, MCP Server, React, Angular, HTML5, CSS3 (Sass), Bootstrap, Material Design, Linux CLI, Git, Remote Model Training, Inference Hosting

## Skills

Platform & Data Engineering:
Docker, Terraform, Kubernetes, Model Deployment, PostgreSQL, MongoDB, NoSQL, Hive, Hadoop, PySpark, Kafka, Feast (Feature Store), DVC (Data Version Control), MLflow, Airflow, AWS SageMaker, AWS ECR, AWS ECS, OpenTelemetry, Grafana

Cloud: Amazon Web Services (AWS), Google Cloud Platform (GCP), Microsoft Azure

Platform: Linux, Window, MacOS

Software Tools: GitHub, Jira, VIm, JetBrain IDEs, Visual Studio Code, Postman, Docker

## Projects

### E-Mind: Translating Slides into Interactive Lecture Videos
01/2024 - 12/2024

Pitched E-Mind: an AI platform using LlamaIndex and LLMs to turn slides into interactive lecture videos.

- Increased script generation efficiency by 75% using LLMs via LlamaIndex for accurate and engaging narration
- Enhanced educational engagement by 50% by implementing a solution bridging static and interactive learning materials
- Transforms static presentation slides into interactive, narrated lecture videos by orchestrating LlamaIndex pipelines, custom prompting, and LLM agents
- Design, build, fine-tune Hugging Face language models on a custom Vietnamese educational dataset, and deploy AI model on AWS cloud with scalable backend API

## Publications

### SOICT '23 Conference
12/2023

– Enhancing Video Retrieval with Robust CLIP-Based Multimodal System | Proceedings of the 12th International Symposium on Information and Communication Technology

Enhacing Video Retrival with Robust CLIP-based Multimodal System

*Dung Le/Tuan Quang*

Co-authored a peer-reviewed publication for the **SOICT '23 Conference** detailing a CLIP-based multimodal system designed to enhance video retrieval accuracy. I engineered the end-to-end pipeline—from **PyTorch** model fine-tuning to a high-performance **FastAPI** backend—resulting in a robust, stakeholder-vetted solution for aligning visual and textual data.

### Computer Vision Foundation
10/2024

– [2410.20883] Improving Generalization in Visual Reasoning via Self-Ensemble

Improving Generalization in Visual Reasoning via Self-Ensemble

*Tien Huy Nguyen/Tuan Quang*

Co-authored research on "Improving Generalization in Visual Reasoning via Self Ensemble," published on **arXiv**, which introduced a training-free framework to enhance the reasoning capabilities of **Large Vision-Language Models (LVLMs)**. I designed and implemented a self-ensemble method that unlocks latent model potential without additional parameter updates, achieving **state-of-the-art (SOTA)** results across benchmarks like SketchyVQA, Outside Knowledge VQA, and out-of- distribution VQA tasks. To support these findings, I managed the full experimental lifecycle in an **Ubuntu** environment, optimizing computational efficiency and system resources for large-scale model evaluation.

## Education

### California State University of Northridge
Los Angeles, United States

Bachelor of Science in Computer Science/Engineering
08/2015 - 08/2019

- Computer Engineering Bachelor of Science Degree - Cum Laude
- **GPA: 3.67**
- **Outstanding Graduating Senior Award, CSUN** (2019)
- **University Scholarship, CSUN** (2018)